

This draft has not been peer-reviewed. The proof is computer-assisted. The new results of Sections 3 to 6 have passed numerical tests, but no one outside this project has checked them, and nothing here is formalised in Lean.

THE UNDECANOPE: AN 88-VERTEX CONVEX POLYHEDRON WITHOUT RUPERT’S PROPERTY

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ABSTRACT. A convex polyhedron is Rupert if a hole can be cut through it that lets a congruent copy pass. Steininger and Yurkevich constructed the Noperthedron, a 90-vertex polyhedron that is not Rupert, with rotation group C_{15} and point symmetry. We construct a second solid of this kind, the Undecanope. It has 88 vertices, rotation group C_{11} and point symmetry, and we prove that it is not Rupert. Near poses where the two shadows coincide, the published method needs three shadow vertices with a radial property. The Undecanope has a band of views where no such triple exists. We prove a local theorem that uses every shadow vertex with both of its edge normals instead. It has a first-order form and a second-order form, and both have exact rational versions. An exact replay of a cover of the pose space by 3.4 million boxes completes the proof. We also show that a negative first-order depth at a generic view makes a point-symmetric solid locally Rupert.

1. INTRODUCTION

John Wallis proved a claim of Prince Rupert of the Rhine: a hole can be cut through a cube that lets a second cube of the same size pass through [Wal85]. A convex body with this property is called *Rupert*. All five Platonic solids are Rupert [Scr68, JWY17], and Jerrard, Wetzel and Yuan asked whether every convex polyhedron is Rupert [JWY17]. Many Archimedean, Catalan and Johnson solids have since been shown to be Rupert [CYZ18, Hof19, Lav19, Fre22, SY23]. Steininger and Yurkevich survey this work in [SY23].

In 2025 Steininger and Yurkevich answered the question [SY25]. Their *Noperthedron* is the union of the orbits of three points under the group of order 30 generated by the rotation $R_z(2\pi/15)$ and the point reflection $-\text{Id}$, so its rotation group is C_{15} . It has 90 vertices, and it is not Rupert. Their proof divides a five-dimensional space of poses into about 18 million boxes. It excludes each box with a *global theorem* or a *local theorem*, in exact rational versions that a computer checks. Reed and Renshaw have since formalised the whole proof in Lean [RR]. Their formalisation includes a second-order form of the global theorem with a separate radius for each angle, which we use.

Zeng studied a family of eight-vertex stellated tetrahedra and gave evidence that some of them are not Rupert [Zen26]. The Omniscience Project has announced a computer-assisted proof that one of them, $P_{11/20}$, is not Rupert [Omn26]. We have not checked that proof, and the announcement was no longer online on 6 October 2026.

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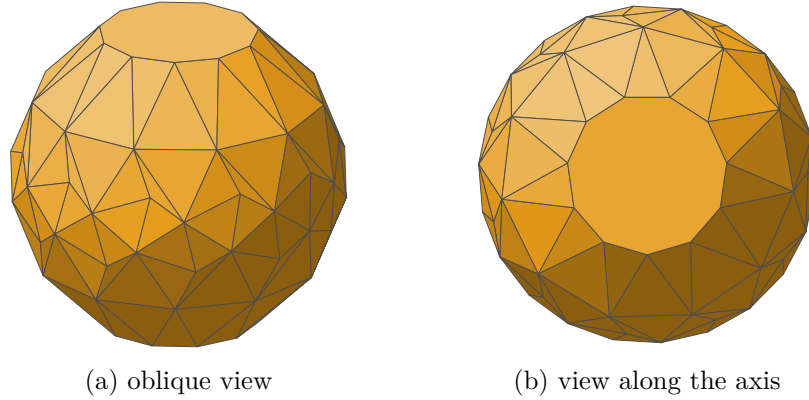


FIGURE 1. The Undecanope. (a) From the view $(\theta, \varphi) = (0.62, 1.18)$. (b) Along the z -axis, where the undecagon of the orbit of g_1 faces the viewer. The vertices are the $\mathcal{C}_{22}$ -orbits of the four generators of Definition 2.2.

The Undecanope. We construct a second polyhedron of the Noperthedron's kind. Let $\mathcal{C}_{22}$ be the group of order 22 generated by $R_z(2\pi/11)$ and $-\text{Id}$. The *Undecanope* $\mathcal{U}$ is the union of the $\mathcal{C}_{22}$ -orbits of the four points $g_1, \dots, g_4$ of Definition 2.2. Its name joins *undecim*, Latin for eleven, to the *nope* of the Noperthedron. It has 88 vertices, 242 edges and 156 faces: 154 triangles and two regular undecagons (Figure 1).

Theorem 1.1. *The Undecanope does not have Rupert's property.*

The proof follows the plan of [SY25] and adds one new ingredient. Call a pose *diagonal* if its two shadows coincide. Near a diagonal pose the global theorem cannot work, because every inner vertex lies close to the outer shadow. Steininger and Yurkevich treat these poses with their local theorem [SY25, Theorem 36]. For small boxes it needs three front-facing shadow vertices that surround the origin, each farther out than every other shadow point in its own direction. We call this condition the *radial gate* (Section 3.3). The Noperthedron was chosen so that the gate holds at every view [SY25, Section 9.1]. The Undecanope fails it at 150 of 65,536 sampled views, which fall in five small clusters (Figure 2).

A local theorem without the radial gate. At a view v , each shadow vertex contributes one vector in $\mathbb{R}^3$ for each of its two edges (Definition 3.3). The vector records how fast that vertex crosses the line of that edge when the inner copy turns. Let $\mu(v)$ be the depth of these vectors. When the origin lies inside their convex hull, $\mu(v)$ is the radius of the largest ball about the origin inside the hull. Otherwise it is minus the distance from the origin to the hull. A view is *generic* if each vertex of its shadow is the image of one vertex of the solid, and no other projected vertex lies on the boundary of the shadow.

Theorem 1.2. *Let $\mathcal{P}$ be a point-symmetric convex polyhedron, and let $v = (\theta, \varphi)$ be a generic view with $0 < \theta < 2\pi$ and $0 < \varphi < \pi$.*

- (a) *If $\mu(v) > 0$, then some neighbourhood of the diagonal pose $(\theta, \varphi, \theta, \varphi, 0)$ contains no solution of Rupert's condition.*
- (b) *If $\mu(v) < 0$, then $\mathcal{P}$ is locally Rupert in the sense of [SY25, Definition 52].*

Part (a) is Corollary 4.6, and part (b) is Proposition 3.4. The radial gate implies $\mu(v) > 0$ (Lemma 3.6), so at first order part (a) applies wherever the published local theorem does. For the certificate we need explicit boxes, including boxes at views where a vertex lies on or near

the silhouette. Theorem 4.2 gives such boxes from μ and a support margin. Theorem 5.2 is a second-order form that keeps the exact dependence on the angle of rotation. Section 6 gives rational versions of both, in the form that a computer checks. Together they answer in part the first open problem of [SY25, Section 9.2]: “Are there other ways to disprove the existence of local solutions?”

The certificate. The certificate covers the reduced pose space with 216,000 cells and 3,402,298 boxes. An exact rational replay checked every box in 17.7 processor-hours. Of the 1,809,149 leaves, 36,470 use the new theorems, and the rest use the global and local theorems of [SY25, RR].

Use of AI. AI models did most of this work under the author’s direction, and the acknowledgements say which model did what. In particular, an AI model found and proved the new local theorems. They have passed extensive numerical tests (Section 7.4), but no one outside this project has reviewed them yet. The code, the certificate and the tests will be made public [Data], so that others can check them.

The author’s Non-Rupert Polyhedra Explorer at <https://drbuild.uk/app/nonrup/> shows the Undecanope and the Noperthedron. It cuts a hole the shape of a solid’s shadow, and lets the reader try to push a copy through.

Organisation. Section 2 defines the Undecanope and reduces the pose space by its symmetries. Section 3 sets up the first-order analysis near diagonal poses, proves Proposition 3.4, and compares μ with the radial gate. Sections 4 and 5 prove the gap and patch forms of the new local theorem, and Section 6 gives their rational versions. Section 7 describes the certificate, its replay and its tests. Section 8 lists open questions.

2. THE UNDECANOPE

2.1. Rupert’s property. We use the notation of [SY25]. For $\alpha, \theta, \varphi \in \mathbb{R}$ let

$$R(\alpha) = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix}, \quad X(\theta, \varphi) = \begin{pmatrix} \cos \theta \sin \varphi \\ \sin \theta \sin \varphi \\ \cos \varphi \end{pmatrix},$$

$$M(\theta, \varphi) = \begin{pmatrix} -\sin \theta & \cos \theta & 0 \\ -\cos \theta \cos \varphi & -\sin \theta \cos \varphi & \sin \varphi \end{pmatrix}.$$

The matrix $M(\theta, \varphi)$ projects $\mathbb{R}^3$ onto $\mathbb{R}^2$ along $X(\theta, \varphi)$, and we call (θ, φ) a *view*. A *polyhedron* is a finite set $\mathcal{P} \subset \mathbb{R}^3$ of points in convex position, not all in one plane. For a finite set $\mathcal{Q} \subset \mathbb{R}^2$ we write $\mathcal{Q}^\circ$ for the interior of its convex hull. A polyhedron is *point-symmetric* if $\mathcal{P} = -\mathcal{P}$. Norms of vectors are Euclidean, and norms of matrices are operator norms.

Definition 2.1 ([SY25, Definition 3]). A point-symmetric polyhedron $\mathcal{P}$ is *Rupert* if some pose $\Psi = (\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha)$ with $\theta_1, \theta_2 \in [0, 2\pi)$, $\varphi_1, \varphi_2 \in [0, \pi]$ and $\alpha \in [-\pi, \pi)$ satisfies

$$(1) \quad R(\alpha)M(\theta_1, \varphi_1)\mathcal{P} \subset (M(\theta_2, \varphi_2)\mathcal{P})^\circ.$$

Such a pose is a *solution*. We call $M(\theta_2, \varphi_2)\mathcal{P}$ the *outer shadow* and $R(\alpha)M(\theta_1, \varphi_1)\mathcal{P}$ the *inner shadow*. For point-symmetric polyhedra, Definition 2.1 agrees with the usual definition, which also allows a translation [SY23, Proposition 2]. A point-symmetric polyhedron is *locally Rupert* if for every $\varepsilon > 0$ it has a solution with $|\theta_1 - \theta_2|, |\varphi_1 - \varphi_2|, |\alpha| \leq \varepsilon$ [SY25, Definition 52]. Scott introduced the same notion [Sco22].

2.2. Definition and symmetries. Let $R_z(\alpha)$ be the rotation by α about the z -axis, as in [SY25, Definition 5], and let

$$C_{22} = \{(-1)^\ell R_z(2\pi k/11) : 0 \leq k \leq 10, \ell \in \{0, 1\}\}.$$

This is a cyclic group of order 22, generated by $-R_z(2\pi/11)$. Its rotations form the cyclic group C_{11} . Its other eleven elements are roto reflections: a rotation about the z -axis followed by the reflection in the plane $z = 0$. In Schoenflies notation it is the group S_{22} .

Definition 2.2. Let

$$\begin{aligned} g_1 &= (0.46216927401999314, 0, 0.8867917242238069), \\ g_2 &= (0.8073903514907679, 0.26028340170707587, 0.4895982440781035), \\ g_3 &= (0.8349899144334889, 0.3966546584642011, 0.22932860156832968), \\ g_4 &= (0.9677138509699689, 0.07927551684849105, 0.09931730273340919), \end{aligned}$$

where each decimal is exact. The *Undecanope* is $\mathcal{U} = C_{22}g_1 \cup C_{22}g_2 \cup C_{22}g_3 \cup C_{22}g_4$.

The 88 points of $\mathcal{U}$ are distinct and in convex position, so $\mathcal{U}$ is a point-symmetric polyhedron with 88 vertices. Its convex hull has 242 edges and 156 faces. Two faces are regular undecagons, the orbits of g_1 and $-g_1$ under C_{11} , and the other 154 faces are triangles. The norms of g_2, g_3 and g_4 are about 0.9795, 0.9524 and 0.9760, and $1 - |g_1|^2 \approx 1.05 \times 10^{-16}$, so every vertex has norm less than 1. The theorems of [SY25] assume radius exactly 1. Their proofs use only the bound $\|P\| \leq 1$ for every vertex P , and the Lean formalisation states the hypothesis in that form [RR].

A floating-point search found $g_1, \dots, g_4$, and we take their decimal forms as exact. The group C_{22} acts on $\mathcal{U}$. A numerical check finds no other symmetries, and we do not use that check.

Lemma 2.3. *For all $\theta, \varphi, \alpha \in \mathbb{R}$ the following hold.*

- (a) $M(\theta + 2\pi/11, \varphi)\mathcal{U} = M(\theta, \varphi)\mathcal{U}$.
- (b) $R(\alpha + \pi)M(\theta, \varphi)\mathcal{U} = R(\alpha)M(\theta, \varphi)\mathcal{U}$.
- (c) $M(\theta + \pi, \pi - \varphi) = D M(\theta, \varphi)$ as matrices, where $D = \text{diag}(-1, 1)$.

Proof. A direct computation gives $M(\theta + \delta, \varphi) = M(\theta, \varphi)R_z(-\delta)$ for every $\delta \in \mathbb{R}$. Part (a) follows, since $R_z(-2\pi/11)\mathcal{U} = \mathcal{U}$. Part (b) follows from $R(\pi) = -\text{Id}$ and $\mathcal{U} = -\mathcal{U}$. For part (c), the first row of $M(\theta + \pi, \pi - \varphi)$ is $(\sin \theta, -\cos \theta, 0)$, the negative of the first row of $M(\theta, \varphi)$. Its second row is $(-\cos \theta \cos \varphi, -\sin \theta \cos \varphi, \sin \varphi)$, the same as the second row of $M(\theta, \varphi)$. $\square$

Part (c) holds for every polyhedron. The third identity of [SY25, Lemma 7] plays the same role, but it uses the symmetry of the Noperthedron.

Corollary 2.4. *If $\mathcal{U}$ is Rupert, then Rupert's condition (1) for $\mathcal{U}$ has a solution in*

$$\Theta = [0, 2\pi/11] \times [0, \pi] \times [0, 2\pi/11] \times [0, \pi/2] \times [-\pi/2, \pi/2].$$

Hence it has a solution in the root box

$$B_0 = [0, \frac{143}{250}] \times [0, \frac{63}{20}] \times [0, \frac{143}{250}] \times [0, \frac{79}{50}] \times [-\frac{79}{50}, \frac{79}{50}].$$

Proof. Let Ψ be a solution, and apply D to both sides of (1). Since $DR(\alpha) = R(-\alpha)D$, Lemma 2.3(c) shows that $(\theta_1 + \pi, \pi - \varphi_1, \theta_2 + \pi, \pi - \varphi_2, -\alpha)$ also satisfies (1). So we may assume that $\varphi_2 \leq \pi/2$. Parts (a) and (b) of Lemma 2.3 then move θ_1 and θ_2 into $[0, 2\pi/11]$ and α into $[-\pi/2, \pi/2]$, without changing either shadow. Finally $\Theta \subset B_0$, because $2\pi/11 < 0.5712 < 143/250$, $\pi < 63/20$ and $\pi/2 < 79/50$. The checker confirms these inequalities with a rational enclosure of π from Machin's formula. $\square$

This is [SY25, Corollary 8] with 15 replaced by 11.

3. LOCAL ANALYSIS NEAR DIAGONAL POSES

3.1. Frames. Let $J = R(\pi/2)$, and let $\text{pr}: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ be the projection $(x, y, z) \mapsto (x, y)$. Then $\text{pr } R_z(\alpha) = R(\alpha) \text{pr}$ for every α . Let $A(\theta, \varphi)$ be the 3×3 matrix whose first two rows are the rows of $M(\theta, \varphi)$ and whose third row is $X(\theta, \varphi)^t$. By the identities of [SY25, Section 2.2],

$$A(\theta, \varphi) = R_z(-\pi/2) R_y(\varphi) R_z(-\theta),$$

with R_y as in [SY25, Definition 5]. So $A(\theta, \varphi)$ is a rotation, and $M(\theta, \varphi) = \text{pr } A(\theta, \varphi)$.

For a pose $\Psi = (\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha)$ write $A_i = A(\theta_i, \varphi_i)$ and $M_i = M(\theta_i, \varphi_i)$, and put

$$W(\Psi) = R_z(\alpha) A_1 A_2^t.$$

Then $\text{pr } W(\Psi) A_2 = \text{pr } R_z(\alpha) A_1 = R(\alpha) M_1$, that is,

$$(2) \quad R(\alpha) M(\theta_1, \varphi_1) = \text{pr } W(\Psi) A(\theta_2, \varphi_2).$$

For a vertex P_k let $x_k = A_2 P_k = (q_k, \eta_k)$, with $q_k = M_2 P_k$ and $\eta_k = \langle X(\theta_2, \varphi_2), P_k \rangle$. These are the coordinates of P_k in the frame of the outer view, and η_k is its height towards the viewer. By (2), inner vertex k is $\text{pr } W(\Psi) x_k$, and outer vertex k is $q_k = \text{pr } x_k$. The pose is *diagonal* if $W(\Psi) = \text{Id}$, and then the two shadows coincide.

For a rotation W let $\psi(W) \in [0, \pi]$ be its angle, so that $1 + 2 \cos \psi(W) = \text{tr } W$. A rotation vector of W is a vector $\omega \in \mathbb{R}^3$ with $|\omega| = \psi(W)$ and $W = \exp[\omega]_\times$, where $[\omega]_\times y = \omega \times y$. When $\psi(W) = \pi$ there are two choices, and either will do below. The quantity $\psi(U^t V)$ is the geodesic distance between rotations U and V for the bi-invariant metric on $SO(3)$, so it satisfies the triangle inequality, and $\psi(UVU^t) = \psi(V)$.

Lemma 3.1. *Let $\theta, \theta', \varphi, \varphi' \in \mathbb{R}$ with $|\theta - \theta'| \leq 2$ and $|\varphi - \varphi'| \leq 2$, and let Ψ be a pose with $|\alpha| \leq \pi$.*

- (a) $\psi(A(\theta, \varphi) A(\theta', \varphi')^t) \leq \sqrt{(\theta - \theta')^2 + (\varphi - \varphi')^2}$.
- (b) $\|A(\theta, \varphi) - A(\theta', \varphi')\| \leq \psi(A(\theta, \varphi) A(\theta', \varphi')^t)$.
- (c) $\psi(W(\Psi)) \leq |\alpha| + \psi(A_1 A_2^t)$.

Proof. (a) Put $a = \theta' - \theta$ and $b = \varphi - \varphi'$. Since $R_y(\varphi')^t = R_y(-\varphi') = R_y(b) R_y(\varphi)^t$, we have

$$A(\theta, \varphi) A(\theta', \varphi')^t = R_z(-\frac{\pi}{2}) R_y(\varphi) (R_z(a) R_y(b)) R_y(\varphi)^t R_z(-\frac{\pi}{2})^t,$$

which is conjugate to $R_z(a) R_y(b)$. The diagonal entries of $R_z(a) R_y(b)$ are $\cos a \cos b$, $\cos a$ and $\cos b$. So the angle ψ of the product satisfies $1 + 2 \cos \psi = \cos a + \cos b + \cos a \cos b$, that is, $2 + 2 \cos \psi = (1 + \cos a)(1 + \cos b)$. By [SY25, Lemma 11], which needs $|a|, |b| \leq 2$, the right side is at least $2 + 2 \cos \sqrt{a^2 + b^2}$. Hence $\cos \psi \geq \cos \sqrt{a^2 + b^2}$. Both ψ and $\sqrt{a^2 + b^2} \leq 2\sqrt{2}$ lie in $[0, \pi]$, where cosine decreases, so $\psi \leq \sqrt{a^2 + b^2}$.

(b) Right multiplication by a rotation preserves the operator norm, so $\|A - A'\| = \|AA'^t - \text{Id}\|$. For a rotation V by angle ψ , the eigenvalues of $V - \text{Id}$ are 0 and $e^{\pm i\psi} - 1$, and $V - \text{Id}$ is normal. So $\|V - \text{Id}\| = |e^{i\psi} - 1| = 2 \sin(\psi/2) \leq \psi$.

(c) By the triangle inequality for $\psi(U^t V)$,

$$\psi(W) \leq \psi(A_1 A_2^t) + \psi((A_1 A_2^t)^t R_z(\alpha) A_1 A_2^t) = \psi(A_1 A_2^t) + \psi(R_z(\alpha)),$$

and $\psi(R_z(\alpha)) = |\alpha|$ for $|\alpha| \leq \pi$. □

Lemma 3.2. *Let $W = \exp[\omega]_\times$ with $\psi = |\omega|$, and let $x = (q, \eta) \in \mathbb{R}^2 \times \mathbb{R}$. Then*

$$\text{pr}(Wx) - q = \omega_3 Jq + \eta(\omega_2, -\omega_1) + r, \quad |r| \leq \frac{1}{2} \psi^2 |x|.$$

Proof. For $\psi = 0$ there is nothing to prove. Otherwise let $\hat{\omega} = \omega/\psi$. Rodrigues' formula $Wx = x + \sin \psi (\hat{\omega} \times x) + (1 - \cos \psi) \hat{\omega} \times (\hat{\omega} \times x)$ gives

$$Wx - x - \omega \times x = (\sin \psi - \psi) \hat{\omega} \times x + (1 - \cos \psi) \hat{\omega} \times (\hat{\omega} \times x).$$

The two vectors on the right are orthogonal, and each has length $|x_\perp|$, where $x_\perp$ is the part of x orthogonal to $\hat{\omega}$. So $|Wx - x - \omega \times x| = |x_\perp| |e^{i\psi} - 1 - i\psi|$, and

$$|e^{i\psi} - 1 - i\psi| = \left| \int_0^\psi i(e^{it} - 1) dt \right| \leq \int_0^\psi t dt = \frac{1}{2}\psi^2.$$

Finally $\text{pr}(x) = q$, and expanding the cross product gives $\text{pr}(\omega \times x) = \omega_3 Jq + \eta(\omega_2, -\omega_1)$. Since pr does not increase lengths, the remainder $r = \text{pr}(Wx - x - \omega \times x)$ satisfies the bound. $\square$

For a unit vector $n \in \mathbb{R}^2$ define the linear map $L_n: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ by

$$L_n(q, \eta) = (\langle Jq, n \rangle, \eta n),$$

and for $\omega \in \mathbb{R}^3$ put $m(\omega) = (\omega_3, \omega_2, -\omega_1)$. Then $|L_n x| \leq |x|$, $|m(\omega)| = |\omega|$, and Lemma 3.2 reads

$$(3) \quad \langle \text{pr}(Wx) - q, n \rangle = \langle L_n x, m(\omega) \rangle + \langle r, n \rangle, \quad |r| \leq \frac{1}{2}|\omega|^2|x|.$$

For a finite set $U \subset \mathbb{R}^3$ define

$$\text{depth}(U) = \min_{|m|=1} \max_{u \in U} \langle u, m \rangle.$$

We use four standard facts about support functions. If $\mu > 0$, then $\text{depth}(U) \geq \mu$ if and only if the closed ball of radius μ about the origin lies in $\text{conv } U$. By separation, $\text{depth}(U) < 0$ if and only if $0 \notin \text{conv } U$, and then $-\text{depth}(U)$ is the distance from 0 to $\text{conv } U$. Adding points to U does not decrease its depth. Finally, if $U = \{u_1, \dots, u_p\}$ and $U' = \{u'_1, \dots, u'_p\}$, then $|\text{depth}(U) - \text{depth}(U')| \leq \max_i |u_i - u'_i|$.

3.2. First-order vectors and passages.

Definition 3.3. Let $\mathcal{P} = \{P_1, \dots, P_N\}$ be a polyhedron, and let $v = (\theta, \varphi)$ be a view. Write $q_k = M(v)P_k$, $\eta_k = \langle X(v), P_k \rangle$ and $x_k = A(v)P_k = (q_k, \eta_k)$, and let K be the convex hull of the q_k . The view is *generic* if every vertex of K equals q_k for exactly one index k , and every other point q_j lies in the interior of K . At a generic view, let n_k^- and n_k^+ be the outward unit normals of the two edges of K at a vertex q_k . The *first-order vectors* at v are

$$U(v) = \{L_n x_k : q_k \text{ a vertex of } K, n \in \{n_k^-, n_k^+\}\},$$

and the *first-order depth* is $\mu(v) = \text{depth } U(v)$.

By (3), when the inner copy turns by $\exp(t[\omega]_\times)$ from the diagonal pose at v , inner vertex k crosses the line of the edge with normal n at rate $\langle L_n x_k, m(\omega) \rangle$ per unit of t . A negative $\mu(v)$ names a turn that moves every shadow vertex inwards across both of its edges. A positive $\mu(v)$ says that every turn moves some shadow vertex outwards, at a rate of at least $\mu(v)|\omega|$.

Proposition 3.4. Let $\mathcal{P}$ be a point-symmetric polyhedron, and let $v = (\bar{\theta}, \bar{\varphi})$ be a generic view with $0 < \bar{\theta} < 2\pi$ and $0 < \bar{\varphi} < \pi$. If $\mu(v) < 0$, then $\mathcal{P}$ is locally Rupert.

Proof. Choose a unit vector m with $\langle u, m \rangle < 0$ for every $u \in U(v)$. Let $\omega = (-m_3, m_2, m_1)$, so that $m(\omega) = m$, and put $W_t = \exp(t[\omega]_\times)$ for $t > 0$. Let K be the outer shadow at v . It has non-empty interior, because $\mathcal{P}$ is not planar. For each edge e of K let n_e be its outward unit normal and $h_e = \max_j \langle q_j, n_e \rangle$ its support value. The interior of K is the set of points y with $\langle y, n_e \rangle < h_e$ for every edge e .

Fix an edge e and an index j . If $\langle q_j, n_e \rangle < h_e$, then $\langle \text{pr}(W_t x_j), n_e \rangle < h_e$ for all small t , by continuity. If $\langle q_j, n_e \rangle = h_e$, then q_j lies on e , so by genericity it is an endpoint of e , and n_e is one of its two edge normals. By (3),

$$\langle \text{pr}(W_t x_j), n_e \rangle - h_e \leq t \langle L_{n_e} x_j, m \rangle + \frac{1}{2} t^2 |x_j|,$$

which is negative for all small $t > 0$. There are finitely many pairs (j, e) , so for all small $t > 0$ every point $\text{pr}(W_t x_j)$ lies in the interior of K .

It remains to realise $W_t A(v)$ by a pose. With the convention of [SY25, Definition 5],

$$R_z(\alpha) A(\theta, \varphi) = R_z(\alpha - \pi/2) R_y(\varphi) R_z(-\theta).$$

This is an Euler-angle chart of $SO(3)$, and it is a local diffeomorphism at $(\bar{\theta}, \bar{\varphi}, 0)$ because $\sin \bar{\varphi} \neq 0$. So for small t there are angles $(\theta_1, \varphi_1, \alpha)$, tending to $(\bar{\theta}, \bar{\varphi}, 0)$ as $t \rightarrow 0$, with $R_z(\alpha) A(\theta_1, \varphi_1) = W_t A(v)$. The pose $\Psi_t = (\theta_1, \varphi_1, \bar{\theta}, \bar{\varphi}, \alpha)$ has $W(\Psi_t) = W_t$, and it lies in the ranges of Definition 2.1 for small t . By (2) its inner shadow is $\{\text{pr}(W_t x_j)\}$, which lies in the interior of K . So Ψ_t is a solution, and $|\theta_1 - \bar{\theta}|, |\varphi_1 - \bar{\varphi}|, |\alpha| \rightarrow 0$ as $t \rightarrow 0$. $\square$

Proposition 3.4 gives a first-order test for local Rupertness. Compare Scott [Sco22], who gives two sufficient conditions for a polyhedron to be locally Rupert.

3.3. The radial gate.

Definition 3.5. Let v be a view, with q_k and η_k as in Definition 3.3. An index k is *eligible* at v if $\eta_k > 0$ and $\langle q_k, q_k - q_j \rangle > 0$ for every $j \neq k$. The *radial gate* holds at v if there are three eligible indices k_1, k_2, k_3 with $\langle Jq_{k_i}, q_{k_{i+1}} \rangle > 0$ for $i = 1, 2, 3$, where $k_4 = k_1$.

By [SY25, Lemma 26], the last condition puts the origin strictly inside the triangle $q_{k_1} q_{k_2} q_{k_3}$. The radial gate is the limit as $\varepsilon \rightarrow 0$ of the hypotheses of [SY25, Theorem 36] with $P_i = Q_i$ and $\sigma_P = \sigma_Q = 0$. Condition (A_ε) becomes $\eta_k > 0$, the ε -spanning condition becomes the condition on the triangle, and condition (B_ε) becomes $\langle q_k, q_k - q_j \rangle > 0$. The search used the radial gate to screen views.

Lemma 3.6. *If the radial gate holds at a generic view v , then $\mu(v) > 0$.*

Proof. Let k be eligible. Then q_k is the unique maximiser of $\langle \cdot, q_k \rangle$ over the shadow, so q_k is a vertex of K and the direction of q_k lies in the interior of its normal cone. That cone is spanned by n_k^- and n_k^+ and has angle less than π , so $q_k = a n_k^- + b n_k^+$ with $a, b > 0$. The map $n \mapsto L_n x_k$ is linear, and $\langle Jq_k, q_k \rangle = 0$, so

$$(0, \eta_k q_k) = L_{q_k} x_k = a L_{n_k^-} x_k + b L_{n_k^+} x_k.$$

Hence $\text{conv } U(v)$ contains $c_k(0, \eta_k q_k)$ with $c_k = 1/(a + b) > 0$. The first coordinates $\langle Jq_k, n_k^\pm \rangle$ of $L_{n_k^\pm} x_k$ satisfy $a \langle Jq_k, n_k^- \rangle + b \langle Jq_k, n_k^+ \rangle = 0$. If both vanished, n_k^- and n_k^+ would both be parallel to q_k , and the cone they span would have angle 0 or π . So one is positive and the other negative.

Now take the three eligible indices of the radial gate. The points $c_k(0, \eta_k q_k)$ lie in the plane $\{u_1 = 0\}$, and since $\eta_k > 0$ their triangle contains a disc about the origin in that plane. The set $\text{conv } U(v)$ contains this disc and points on both sides of the plane, so it contains a ball about the origin. Hence $\mu(v) > 0$. $\square$

The converse fails. Table 1 compares the radial gate with the sign of μ on 65,536 fresh views. The Undecanope fails the gate at 137 of them, yet μ stays positive at every sampled view. Over the 150 gate failures of Figure 2, the smallest value of μ , scaled as in the table, is 7.4×10^{-4} .

Solid	Views with $\mu < 0$	Gate failures	Least μ
Noperthedron	0	0	+0.00884
52-vertex C_{13} control	2,204	10,832	-0.0124
Undecanope	0	137	+0.00146

TABLE 1. The radial gate and the first-order depth on 65,536 fresh views (seed 20261100), in floating point. Each vector of $U(v)$ was divided by the support value of its edge. This changes the value of μ but not its sign. The 52-vertex control has two generator orbits and is Rupert.

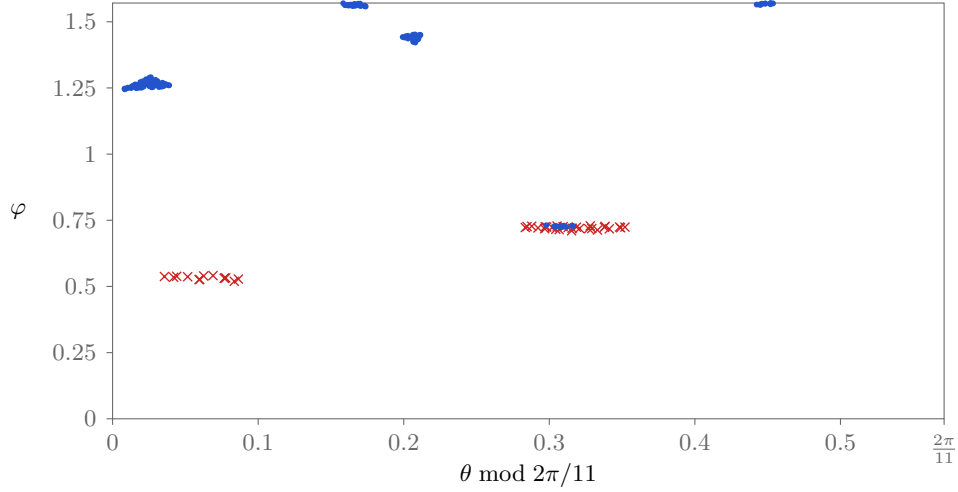


FIGURE 2. The reduced view domain $0 \leq \theta < 2\pi/11$, $0 \leq \varphi \leq \pi/2$ of the Undecanope. Blue dots: the 150 views, out of 65,536 sampled (seed 20261016), where the radial gate fails. Red crosses: the 40 views, out of 16,384 Sobol views, where the new theorem gives the smallest diagonal boxes. Views are folded into the domain by Lemma 2.3(a) and by the map $(\theta, \varphi) \mapsto (\theta + \pi, \pi - \varphi)$, which mirrors the shadow and so keeps the gate's verdict. One cluster of gate failures lies inside the band of hardest views, near $(0.31, 0.73)$.

4. A LOCAL THEOREM WITHOUT THE RADIAL GATE

In this section $\mathcal{P} = \{P_1, \dots, P_N\}$ is any finite set of points with $\|P_k\| \leq 1$. A *box* is a product of five closed intervals. For a pose Ψ write $x_k(\Psi) = A(\theta_2, \varphi_2)P_k = (q_k(\Psi), \eta_k(\Psi))$, and let $h_\Psi(n) = \max_j \langle q_j(\Psi), n \rangle$ be the support function of the outer shadow. Let S be a finite set of pairs (k, n) , where k is an index and $n \in \mathbb{R}^2$ is a unit vector.

Definition 4.1. Numbers $e, \psi_{\max} \geq 0$, vectors $\tilde{x}_1, \dots, \tilde{x}_N \in \mathbb{R}^3$ and numbers $G_{kn} \geq 0$ for $(k, n) \in S$ are *drift data* for a box B if every pose $\Psi \in B$ satisfies

- (D1) $\psi(W(\Psi)) \leq \psi_{\max}$;
- (D2) $|x_k(\Psi) - \tilde{x}_k| \leq e$ for every k ;
- (D3) $h_\Psi(n) - \langle q_k(\Psi), n \rangle \leq G_{kn}$ for every $(k, n) \in S$.

We then write $\tilde{u}_{kn} = L_n \tilde{x}_k$.

Condition (D2) bounds how far the outer frame drifts across the box, and G_{kn} is a *gap*: it bounds how far vertex k can sit inside the outer shadow in direction n .

Theorem 4.2 (gap form). *Let $(e, \psi_{\max}, \tilde{x}, G)$ be drift data for a box B . Suppose that every $m \in \mathbb{R}^3$ with $0 < |m| \leq \psi_{\max}$ satisfies*

$$(4) \quad \max_{(k,n) \in S} (\langle \tilde{u}_{kn}, m \rangle - G_{kn}) > e|m| + \frac{1}{2}|m|^2.$$

Then no pose in B satisfies Rupert's condition (1).

Proof. Let $\Psi \in B$ and $W = W(\Psi)$. If $W = \text{Id}$, the two shadows are the same set K , and (1) would put the compact set K inside its own interior. That is impossible for a non-empty compact subset of $\mathbb{R}^2$, whose interior cannot also be closed.

Otherwise $\psi = \psi(W) \in (0, \psi_{\max}]$. Let ω be a rotation vector of W , and put $m = m(\omega)$, so that $|m| = \psi$. Fix $(k, n) \in S$ and write $x_k = x_k(\Psi)$. By (3) and (D3),

$$\langle \text{pr}(Wx_k), n \rangle - h_\Psi(n) = \langle \text{pr}(Wx_k) - q_k, n \rangle - (h_\Psi(n) - \langle q_k, n \rangle) \geq \langle L_n x_k, m \rangle - \frac{1}{2}\psi^2 - G_{kn},$$

using $|x_k| = \|P_k\| \leq 1$. By (D2), $|L_n x_k - \tilde{u}_{kn}| \leq |x_k - \tilde{x}_k| \leq e$, so

$$\langle \text{pr}(Wx_k), n \rangle - h_\Psi(n) \geq \langle \tilde{u}_{kn}, m \rangle - e|m| - \frac{1}{2}|m|^2 - G_{kn}.$$

By (4) the right side is positive for some $(k, n) \in S$. For that pair, the inner vertex $\text{pr}(Wx_k) = R(\alpha)M(\theta_1, \varphi_1)P_k$ lies outside the half-plane $\{y : \langle y, n \rangle \leq h_\Psi(n)\}$, which contains the outer shadow. So (1) fails. $\square$

The next lemma supplies drift data from the centre of a box. Here and below $\bar{A} = A(\bar{\theta}, \bar{\varphi})$ has rows $\bar{M}$ and $\bar{X}^t$.

Lemma 4.3. *Let B be a box on which $|\theta_2 - \bar{\theta}| \leq e_\theta$, $|\varphi_2 - \bar{\varphi}| \leq e_\varphi$, $|\theta_1 - \theta_2| \leq d_\theta$, $|\varphi_1 - \varphi_2| \leq d_\varphi$ and $|\alpha| \leq a$, where $e_\theta, e_\varphi, d_\theta, d_\varphi \leq 2$ and $a \leq \pi$. Put $e = \sqrt{e_\theta^2 + e_\varphi^2}$, $\psi_{\max} = a + \sqrt{d_\theta^2 + d_\varphi^2}$, $\tilde{x}_k = \bar{A}P_k$ and*

$$G_{kn} = \max\left(0, \max_{j \neq k} (\langle \bar{M}(P_j - P_k), n \rangle + e\|P_j - P_k\|)\right).$$

Then $(e, \psi_{\max}, \tilde{x}, G)$ are drift data for B , and $\tilde{u}_{kn} = (\langle J\bar{M}P_k, n \rangle, \langle \bar{X}, P_k \rangle n)$.

Proof. Condition (D1) is Lemma 3.1(a) and (c). For (D2) and (D3), let $A_2 = A(\theta_2, \varphi_2)$. Lemma 3.1(a) and (b) give $\|A_2 - \bar{A}\| \leq e$. Hence $|x_k - \tilde{x}_k| \leq \|A_2 - \bar{A}\| \|P_k\| \leq e$, and $\|M_2 - \bar{M}\| \leq \|A_2 - \bar{A}\| \leq e$. For $j \neq k$,

$$\langle q_j - q_k, n \rangle = \langle M_2(P_j - P_k), n \rangle \leq \langle \bar{M}(P_j - P_k), n \rangle + e\|P_j - P_k\| \leq G_{kn},$$

and $h_\Psi(n) - \langle q_k, n \rangle$ is the larger of 0 and these values. $\square$

Condition (4) is a statement about a ball of vectors m . The following corollary reduces it to depths of finite sets, which a computer can certify.

Corollary 4.4 (shells). *Let $(e, \psi_{\max}, \tilde{x}, G)$ be drift data for B . Let $0 = \psi_0 < \psi_1 < \dots < \psi_r = \psi_{\max}$, and let $\tau_1, \dots, \tau_r \geq 0$ with $\tau_1 = 0$. Put $U_i = \{\tilde{u}_{kn} : (k, n) \in S, G_{kn} \leq \tau_i\}$. If*

$$\text{depth}(U_i) > e + \frac{1}{2}\psi_i + \tau_i/\psi_{i-1} \quad (i = 1, \dots, r),$$

where the last term is read as 0 for $i = 1$, then (4) holds, and no pose in B satisfies (1).

Proof. Let $0 < |m| \leq \psi_{\max}$, and choose i with $\psi_{i-1} < |m| \leq \psi_i$. The depth bound gives a pair with $G_{kn} \leq \tau_i$ and $\langle \tilde{u}_{kn}, m \rangle > (e + \frac{1}{2}\psi_i + \tau_i/\psi_{i-1})|m|$. For $i = 1$ this gives $\langle \tilde{u}_{kn}, m \rangle - G_{kn} > e|m| + \frac{1}{2}|m|^2$, since $G_{kn} = 0$ and $|m| \leq \psi_1$. For $i > 1$,

$$\langle \tilde{u}_{kn}, m \rangle - G_{kn} > e|m| + \frac{1}{2}|m|^2 + \tau_i(|m|/\psi_{i-1} - 1) \geq e|m| + \frac{1}{2}|m|^2,$$

since $|m| \leq \psi_i$ and $|m| > \psi_{i-1}$. $\square$

The first shell needs pairs with no gap. Later shells may use vertices that lie on, or just inside, the silhouette, because a larger rotation can push them out by more than their gap. This is what lets the theorem work at views where a vertex crosses the boundary of the shadow.

Corollary 4.5 (diagonal boxes). *Let $0 < \varepsilon \leq 1$, and let B be the cube of half-width ε about the diagonal pose $(\bar{\theta}, \bar{\varphi}, \bar{\theta}, \bar{\varphi}, 0)$. Let*

$$\mu = \text{depth}\{\tilde{u}_{kn} : (k, n) \in S\}, \quad s = \min_{(k,n) \in S} \min_{j \neq k} \frac{\langle \bar{M}(P_k - P_j), n \rangle}{\|P_k - P_j\|}.$$

If $\varepsilon < \min(\mu/(2\sqrt{2} + \frac{1}{2}), s/\sqrt{2})$, then no pose in B satisfies (1).

Proof. Apply Lemma 4.3 with $e_\theta = e_\varphi = \varepsilon$, $d_\theta = d_\varphi = 2\varepsilon$ and $a = \varepsilon$. Then $e = \sqrt{2}\varepsilon$ and $\psi_{\max} = (1 + 2\sqrt{2})\varepsilon$. Since $\sqrt{2}\varepsilon < s$, every gap G_{kn} is zero. Use a single shell: $\mu > (2\sqrt{2} + \frac{1}{2})\varepsilon = e + \frac{1}{2}\psi_{\max}$. $\square$

The ratio s is a directional form of the locally-maximally-distant margin of [SY25, Definition 31]. With the edge normals themselves s is zero, since both endpoints of an edge attain its support value. Normals turned slightly into each normal cone keep s positive, and they change μ only slightly.

Corollary 4.6. *Let $\mathcal{P}$ be a point-symmetric polyhedron, and let v be a generic view with $\mu(v) > 0$. Then some neighbourhood of the diagonal pose $(v, v, 0)$ contains no solution of (1). This proves Theorem 1.2(a).*

Proof. Scaling $\mathcal{P}$ changes neither (1) nor the sign of $\mu(v)$, so we may assume $\|P\| \leq 1$ for all $P \in \mathcal{P}$. For each vertex q_k of the shadow K , let $\hat{n}_k$ be the unit vector along $n_k^- + n_k^+$, and for $0 < t < 1$ let $n_k^\pm(t)$ be the unit vector along $(1-t)n_k^\pm + t\hat{n}_k$. These lie in the interior of the normal cone of K at q_k , so q_k is the unique maximiser of $\langle \cdot, n_k^\pm(t) \rangle$ over K . By genericity no other q_j equals q_k , so the margin s of Corollary 4.5 for the pairs $(k, n_k^\pm(t))$ is positive. As $t \rightarrow 0$ the vectors $L_{n_k^\pm(t)} x_k$ tend to the vectors of $U(v)$, so their depth tends to $\mu(v) > 0$. Fix a small t and apply Corollary 4.5. $\square$

Remark 4.7. The theorem also covers boxes off the diagonal. For the Undecanope, the inner view may first be shifted by whole sectors, since $M(\theta_1 + 2\pi j/11, \varphi_1)\mathcal{U} = M(\theta_1, \varphi_1)\mathcal{U}$; the checker allows $|j| \leq 2$. The region that Theorem 4.2 covers is then a tube of radius about 2μ around the diagonal. The published local theorem [SY25, Theorem 36] compares exact distances instead of first-order rates, and where it applies it covers a tube whose radius is about the locally-maximally-distant margin. That is often wider. The certificate uses whichever theorem closes a box.

The weak point of Theorem 4.2 is the drift e , a single radius for both outer angles. Where a vertex crosses a silhouette edge, its gap changes only across a curve in the outer view, but e charges the full radius in every direction. On the Undecanope this left one region that boxes of practical size could not close. The second-order form below closes it.

5. A SECOND-ORDER PATCH FORM

For a unit vector $\hat{\omega} \in \mathbb{R}^3$, a point $x = (q, \eta) \in \mathbb{R}^2 \times \mathbb{R}$ and a unit vector $n \in \mathbb{R}^2$, put

$$a(\hat{\omega}; x, n) = \langle \text{pr}(\hat{\omega} \times x), n \rangle, \quad b(\hat{\omega}; x, n) = \langle \text{pr}(\hat{\omega} \times (\hat{\omega} \times x)), n \rangle.$$

Lemma 5.1. *Let $\hat{\omega}$, x and n be as above.*

(a) *If $W = \exp(\psi[\hat{\omega}]_\times)$, then $\langle \text{pr}(Wx) - q, n \rangle = \sin \psi a(\hat{\omega}; x, n) + (1 - \cos \psi) b(\hat{\omega}; x, n)$.*

- (b) $a(\hat{\omega}; x, n) = \langle c, \hat{\omega} \rangle$ with $c = (-\eta n_2, \eta n_1, \langle Jq, n \rangle)$, and $b(\hat{\omega}; x, n) = \hat{\omega}^t S \hat{\omega}$ with $S = \frac{1}{2}(x \bar{n}^t + \bar{n} x^t) - \langle q, n \rangle \text{Id}$, where $\bar{n} = (n, 0)$.
(c) For $x, y \in \mathbb{R}^3$, $|a(\hat{\omega}; x, n) - a(\hat{\omega}; y, n)| \leq |x - y|$ and $|b(\hat{\omega}; x, n) - b(\hat{\omega}; y, n)| \leq |x - y|$.

Proof. Part (a) is Rodrigues' formula followed by pr and the inner product with n . For (b), the expansion in the proof of Lemma 3.2 gives $\text{pr}(\hat{\omega} \times x) = \hat{\omega}_3 Jq + \eta(\hat{\omega}_2, -\hat{\omega}_1)$, whose inner product with n is $\langle c, \hat{\omega} \rangle$. Also $\hat{\omega} \times (\hat{\omega} \times x) = \langle \hat{\omega}, x \rangle \hat{\omega} - x$, so $b = \langle \hat{\omega}, \bar{n} \rangle \langle \hat{\omega}, x \rangle - \langle q, n \rangle = \hat{\omega}^t S \hat{\omega}$. Part (c) holds because a and b are linear in x , and $|\hat{\omega} \times y| \leq |y|$ and $|\hat{\omega} \times (\hat{\omega} \times y)| \leq |y|$. $\square$

A cone over vectors $v_1, v_2, v_3 \in \mathbb{R}^3$ is the set $\{\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 : \lambda_i \geq 0\}$.

Theorem 5.2 (patch form). *Let $(e, \psi_{\max}, \tilde{x}, G)$ be drift data for a box B , with $\psi_{\max} \leq \pi/2$. Let finitely many cones cover $\mathbb{R}^3 \setminus \{0\}$. Suppose that for each cone C there is a partition $0 = \psi_0 < \psi_1 < \dots < \psi_r = \psi_{\max}$ and, for each segment $s \in \{1, \dots, r\}$, a pair $(k, n) \in S$ and numbers A_s, B_s with the following properties.*

- (i) $A_s \leq a(\hat{\omega}; \tilde{x}_k, n)$ and $B_s \leq b(\hat{\omega}; \tilde{x}_k, n)$ for every unit vector $\hat{\omega} \in C$.
- (ii) $A'_s = A_s - e > 0$. Put $B'_s = \min(B_s - e, 0)$ and $f_s(\psi) = A'_s \sin \psi + B'_s(1 - \cos \psi) - G_{kn}$.
- (iii) $f_s(\psi_s) > 0$. If $s > 1$ then $f_s(\psi_{s-1}) > 0$, and if $s = 1$ then $G_{kn} = 0$.

Then no pose in B satisfies (1).

Proof. Let $\Psi \in B$ and $W = W(\Psi)$. The case $W = \text{Id}$ is as in Theorem 4.2. Otherwise $W = \exp(\psi[\hat{\omega}]_{\times})$ with $\psi \in (0, \psi_{\max}]$ and $|\hat{\omega}| = 1$. Choose a cone C containing $\hat{\omega}$, a segment s with $\psi_{s-1} \leq \psi \leq \psi_s$, and its pair (k, n) . By Lemma 5.1(a) and (D3),

$$\langle \text{pr}(Wx_k), n \rangle - h_{\Psi}(n) \geq \sin \psi a(\hat{\omega}; x_k, n) + (1 - \cos \psi) b(\hat{\omega}; x_k, n) - G_{kn}.$$

By Lemma 5.1(c), (D2) and (i), $a(\hat{\omega}; x_k, n) \geq A_s - e = A'_s$ and $b(\hat{\omega}; x_k, n) \geq B_s - e \geq B'_s$. Since $\sin \psi \geq 0$ and $1 - \cos \psi \geq 0$, the right side is at least $f_s(\psi)$. On $[0, \pi/2]$, $f_s''(\psi) = -A'_s \sin \psi + B'_s \cos \psi \leq 0$, so f_s is concave there. For $s > 1$, f_s is positive at both ends of the segment and so on all of it. For $s = 1$, $f_1(0) = -G_{kn} = 0$ and $f_1(\psi_1) > 0$, so concavity gives $f_1(\psi) \geq (\psi/\psi_1)f_1(\psi_1) > 0$ for $0 < \psi \leq \psi_1$. In each case the inner vertex k lies outside the outer shadow. $\square$

Lemma 5.3 (cone constants). *Let C be the cone over v_1, v_2, v_3 .*

- (a) If $t \geq 0$ and $\langle c, v_i \rangle \geq t|v_i|$ for $i = 1, 2, 3$, then $\langle c, \hat{\omega} \rangle \geq t$ for every unit vector $\hat{\omega} \in C$.
- (b) Let S be symmetric. If $\langle v_i, v_j \rangle > 0$ and $v_i^t S v_j \geq B \langle v_i, v_j \rangle$ for all i, j , then $\hat{\omega}^t S \hat{\omega} \geq B$ for every unit vector $\hat{\omega} \in C$.

Proof. Write $\omega = \sum_i \lambda_i v_i$ with $\lambda_i \geq 0$. For (a), $\langle c, \omega \rangle \geq t \sum_i \lambda_i |v_i| \geq t|\omega|$. For (b), $\omega^t S \omega - B|\omega|^2 = \sum_{i,j} \lambda_i \lambda_j (v_i^t S v_j - B \langle v_i, v_j \rangle) \geq 0$. $\square$

Lemma 5.4 (cone trees). *The eight cones over $(s_1 e_1, s_2 e_2, s_3 e_3)$, $s_i \in \{\pm 1\}$, cover $\mathbb{R}^3 \setminus \{0\}$. The cone over (v_1, v_2, v_3) is covered by the four cones over (v_1, v_{12}, v_{13}) , (v_{12}, v_2, v_{23}) , (v_{13}, v_{23}, v_3) and (v_{12}, v_{23}, v_{13}) , where $v_{ij} = v_i + v_j$.*

Proof. The first claim is clear. For the second, a cone is unchanged when its generators are scaled. The four cones are the cones over the four triangles that the edge midpoints cut from the triangle $v_1 v_2 v_3$, and those triangles cover it. $\square$

So every finite subdivision tree of the eight octant cones covers $\mathbb{R}^3 \setminus \{0\}$. After one subdivision all generators of a cone have positive inner products, as Lemma 5.3(b) requires.

6. RATIONAL VERSIONS

The certificate checks every leaf with exact rational arithmetic. This section states what the checker verifies for the new leaf types, and why that implies the hypotheses of Sections 4 and 5. It then explains how the checker applies the published theorems.

6.1. Vertices and trigonometry. As in [SY25] we put $\kappa = 10^{-10}$. The checker uses the vertex table $\tilde{P}_k = \lfloor 10^{16} P_k \rfloor / 10^{16}$, rounding each coordinate down, as [SY25, Section 7.2] does for the Noperthedron. An interval computation with a Machin enclosure of π shows that $|P_k - \tilde{P}_k|^2 < 103/(5 \cdot 10^{33}) < \kappa^2$ for every k .

Let $\sin_{\mathbb{Q}}$ and $\cos_{\mathbb{Q}}$ be the Taylor polynomials of degrees 25 and 24 from [SY25, Definition 37]. The checker evaluates them exactly and rounds down to a multiple of 10^{-13} :

$$s(x) = \lfloor 10^{13} \sin_{\mathbb{Q}}(x) \rfloor / 10^{13}, \quad c(x) = \lfloor 10^{13} \cos_{\mathbb{Q}}(x) \rfloor / 10^{13}.$$

Lemma 6.1. *For $|x| \leq 4$, $|s(x) - \sin x|$ and $|c(x) - \cos x|$ are less than $4^{26}/26! + 10^{-13} < 1.13 \times 10^{-11} < \kappa/7$. For $|x| \leq 8/5$ they are less than $10^{-13} + 6 \times 10^{-22}$.*

Proof. The Taylor remainders are at most $|x|^{27}/27!$ and $|x|^{26}/26!$ [SY25, proof of Lemma 38], and rounding adds less than 10^{-13} . Also $(8/5)^{26}/26! < 6 \times 10^{-22}$. $\square$

Let $A_{\mathbb{Q}}(\theta, \varphi)$ be $A(\theta, \varphi)$ with s and c in place of $\sin$ and $\cos$, and let $M_{\mathbb{Q}}$ and $X_{\mathbb{Q}}$ be its rows. Put $\varepsilon_A = 10^{-12}$ and $\Delta = 2 \times 10^{-10}$.

Lemma 6.2. *If $0 \leq \theta, \varphi \leq 8/5$, then $\|A(\theta, \varphi) - A_{\mathbb{Q}}(\theta, \varphi)\| \leq \varepsilon_A$. Hence $|A(\theta, \varphi)P_k - A_{\mathbb{Q}}(\theta, \varphi)\tilde{P}_k| \leq \varepsilon_A + (1 + \varepsilon_A)\kappa \leq \Delta$.*

Proof. Each entry of A is 0, a sine or cosine, or a product of two of them. With $\tau = 10^{-13} + 6 \times 10^{-22}$ from Lemma 6.1, each entry of $A - A_{\mathbb{Q}}$ is at most $\tau + (1 + \tau)\tau < 2.01 \times 10^{-13}$ in absolute value. The Frobenius norm of $A - A_{\mathbb{Q}}$ is then below $3 \times 2.01 \times 10^{-13} < \varepsilon_A$, and the operator norm is no larger. The second claim follows from $\|A_{\mathbb{Q}}\| \leq 1 + \varepsilon_A$, $\|P_k\| \leq 1$ and $|P_k - \tilde{P}_k| \leq \kappa$. $\square$

For a rational $y \geq 0$ let $\lceil \sqrt{y} \rceil$ denote the checker's upper square root, $(\lfloor \sqrt{\lceil 10^{32} y \rceil} \rfloor + 1)/10^{16}$, which is at least $\sqrt{y}$.

Lemma 6.3 (rational drift data). *Let B be a box with rational centre $(\bar{\theta}_1, \bar{\varphi}_1, \bar{\theta}_2, \bar{\varphi}_2, \bar{\alpha})$ and radii $(r_1, \dots, r_5)$, with $0 \leq \bar{\theta}_2, \bar{\varphi}_2 \leq 8/5$, and let $j \in \{-2, \dots, 2\}$. Let $\pi^- \leq \pi \leq \pi^+$ be rational, and put*

$$d_{\theta} = \max(|\bar{\theta}_1 - \bar{\theta}_2 + \frac{2j}{11}\pi^-|, |\bar{\theta}_1 - \bar{\theta}_2 + \frac{2j}{11}\pi^+|) + r_1 + r_3, \quad d_{\varphi} = |\bar{\varphi}_1 - \bar{\varphi}_2| + r_2 + r_4,$$

and suppose $d_{\theta}, d_{\varphi} \leq 2$. Put $e_2 = \lceil \sqrt{r_3^2 + r_4^2} \rceil$, $e = e_2 + \Delta$, $\psi_{\max} = |\bar{\alpha}| + r_5 + \lceil \sqrt{d_{\theta}^2 + d_{\varphi}^2} \rceil$, $\tilde{x}_k = A_{\mathbb{Q}}(\bar{\theta}_2, \bar{\varphi}_2)\tilde{P}_k$, and

$$G_{kn} = \max\left(0, \max_{j' \neq k}(\langle M_{\mathbb{Q}}(\tilde{P}_{j'} - \tilde{P}_k), n \rangle + (e_2 + \varepsilon_A)d_{j'k} + 2\kappa(1 + \varepsilon_A))\right),$$

where $M_{\mathbb{Q}} = M_{\mathbb{Q}}(\bar{\theta}_2, \bar{\varphi}_2)$ and $d_{j'k} = \lceil \sqrt{|\tilde{P}_{j'} - \tilde{P}_k|^2} \rceil + 2\kappa$. Then $(e, \psi_{\max}, \tilde{x}, G)$ are drift data for B , for the Undecanope with its inner view shifted by j sectors.

Proof. Shift each pose by $\theta_1 \mapsto \theta_1 + 2\pi j/11$. This does not change the inner shadow, by Lemma 2.3(a), so Rupert's condition is unchanged. The choice of $\pi^{\pm}$ makes d_{θ} an upper bound for $|\theta_1 + 2\pi j/11 - \theta_2|$ over the box, so (D1) follows from Lemma 3.1(a) and (c). Let

$\bar{A} = A(\bar{\theta}_2, \bar{\varphi}_2)$ and $A_{\mathbb{Q}} = A_{\mathbb{Q}}(\bar{\theta}_2, \bar{\varphi}_2)$. Lemma 3.1 gives $\|A_2 - \bar{A}\| \leq e_2$, and Lemma 6.2 gives $\|\bar{A} - A_{\mathbb{Q}}\| \leq \varepsilon_A$. For (D2),

$$|A_2 P_k - A_{\mathbb{Q}} \tilde{P}_k| \leq \|A_2 - \bar{A}\| + \|\bar{A} P_k - A_{\mathbb{Q}} \tilde{P}_k\| \leq e_2 + \Delta = e.$$

For (D3), $\|M_2 - M_{\mathbb{Q}}\| \leq e_2 + \varepsilon_A$ and $\|M_{\mathbb{Q}}\| \leq 1 + \varepsilon_A$, so for $j' \neq k$,

$$\langle M_2(P_{j'} - P_k), n \rangle \leq \langle M_{\mathbb{Q}}(\tilde{P}_{j'} - \tilde{P}_k), n \rangle + (e_2 + \varepsilon_A)\|P_{j'} - P_k\| + 2\kappa(1 + \varepsilon_A),$$

and $\|P_{j'} - P_k\| \leq d_{j'k}$. $\square$

6.2. Depth certificates. A computer cannot evaluate a depth directly, so the checker accepts a surface as a witness.

Lemma 6.4. *Let $U = \{u_1, \dots, u_p\} \subset \mathbb{R}^3$, and let T be a list of triangles (i, j, l) of distinct indices. Suppose that each ordered pair (i, j) occurs as a directed edge of the triangles of T as often as (j, i) does. Suppose also that $\mu \geq 0$ and that every triangle in T satisfies*

$$\langle N, u_i \rangle > 0 \quad \text{and} \quad \langle N, u_i \rangle^2 > \mu^2 |N|^2, \quad \text{where } N = (u_j - u_i) \times (u_l - u_i).$$

Then $\text{depth}(U) > \mu$. No convexity or embedding of the surface is needed.

Proof. For a point y off the triangles, let $w(y)$ be $1/(4\pi)$ times the sum over T of the signed solid angles of the triangles seen from y . The edge condition makes T a 2-cycle, so $w(y)$ is an integer: it is the degree of the radial projection of T onto a sphere about y . It is locally constant off the triangles. If $y \notin \text{conv } U$, a plane separates y from every triangle, the radial projection misses a hemisphere, and $w(y) = 0$.

The solid angle of the triangle (i, j, l) seen from the origin has the sign of $\det(u_i, u_j, u_l) = \langle N, u_i \rangle$, which is positive. So $w(0) > 0$, and hence $w(0) \geq 1$. Let μ' be the least distance $\langle N, u_i \rangle / |N|$ from the origin to the plane of a triangle. Then $\mu' > \mu$, and the open ball of radius μ' about the origin meets no triangle. On that ball $w = w(0) \neq 0$, so the ball lies in $\text{conv } U$. Hence $\text{depth}(U) \geq \mu' > \mu$. $\square$

The checker requires each directed edge to occur exactly once, together with its reverse, and it rejects a triangle whose vectors are collinear, since then $N = 0$.

6.3. The two new leaf checks.

Theorem 6.5 (rational gap form). *Let B be a box with data as in Lemma 6.3. Suppose a witness gives a finite set S of pairs (k, n) with exact rational unit vectors $n = (n_x, n_y)/d$, $n_x^2 + n_y^2 = d^2$, and shells $0 = \psi_0 < \psi_1 < \dots < \psi_r = \psi_{\max}$ with rational levels $\tau_i \geq 0$ and $\tau_1 = 0$. Suppose that for each i the witness gives triangles on the vectors $\tilde{u}_{kn}$ with $G_{kn} \leq \tau_i$ that satisfy Lemma 6.4 with $\mu_i = e + \frac{1}{2}\psi_i + \tau_i/\psi_{i-1}$ (read $\mu_1 = e + \frac{1}{2}\psi_1$). Then no pose in B satisfies (1) for $\mathcal{U}$.*

Proof. Lemma 6.3 gives drift data, Lemma 6.4 gives the depth bounds of Corollary 4.4, and Theorem 4.2 applies. $\square$

Theorem 6.6 (rational patch form). *Let B be a box with data as in Lemma 6.3 and $\psi_{\max} \leq 1$. Suppose a witness gives pairs (k, n) as in Theorem 6.5, and a subdivision tree of the eight octant cones in which every root is split at least once. For each leaf cone C over (v_1, v_2, v_3) it gives a partition of $[0, \psi_{\max}]$ and a pair for each segment. Let c and S be the vector and matrix of Lemma 5.1(b) for $\tilde{x}_k$ and n , and put*

$$A_C = \min_i \frac{\langle c, v_i \rangle}{\lceil \sqrt{|v_i|^2} \rceil}, \quad B_C = \min_{i,j} \frac{v_i^t S v_j}{\langle v_i, v_j \rangle}.$$

Suppose $\langle c, v_i \rangle > 0$ and $\langle v_i, v_j \rangle > 0$ for all i, j , that $A' = A_C - e > 0$, and, with $B' = \min(B_C - e, 0)$, that

$$F(\psi) = (\psi - \frac{1}{6}\psi^3)A' + \frac{1}{2}\psi^2B' - G_{kn}$$

is positive at the end of each segment and at the start of each segment after the first. Suppose finally that the first segment uses a pair with $G_{kn} = 0$. Then no pose in B satisfies (1) for $\mathcal{U}$.

Proof. Lemma 6.3 gives drift data. Since $A_C > 0$, the definition of A_C gives $\langle c, v_i \rangle \geq A_C \lceil \sqrt{|v_i|^2} \rceil \geq A_C |v_i|$, so Lemma 5.3(a) gives $A_C \leq a(\hat{\omega}; \tilde{x}_k, n)$ on C . Lemma 5.3(b) gives $B_C \leq b(\hat{\omega}; \tilde{x}_k, n)$. For $\psi \geq 0$ we have $\sin \psi \geq \psi - \psi^3/6$ and $1 - \cos \psi \leq \psi^2/2$, and $A' > 0 \geq B'$, so $f_s(\psi) \geq F(\psi)$. Lemma 5.4 shows that the leaf cones cover $\mathbb{R}^3 \setminus \{0\}$, and $\psi_{\max} \leq 1 < \pi/2$. Theorem 5.2 applies. $\square$

6.4. The published theorems. For the other leaves the checker uses the rational global theorem in the second-order form of [RR], with a separate radius for each angle, and the rational local theorem [SY25, Theorem 48]. A global witness is a vertex index and an exact rational unit vector w . A local witness gives the indices of P_1, P_2, P_3 and Q_1, Q_2, Q_3 , a rational r and the sign σ_Q , with $\sigma_P = 0$. The checker confirms that the two triples are congruent through an element of $\mathcal{C}_{22}$. For a triple inside one generator orbit it also accepts the map $k \mapsto k_0 - k$ on rotation indices, which a reflection in a vertical plane induces. It applies the local theorem to the cube with the box's centre and its largest radius.

Lemma 6.7. *The checker evaluates both published theorems with s and c in place of $\sin_{\mathbb{Q}}$ and $\cos_{\mathbb{Q}}$. It rounds down to a multiple of 10^{-13} each entry of the row vectors $w^t R(\bar{\alpha}) M^{(\cdot)}$ and $w^t M^{(\cdot)}$ of the global theorem, and of the projected vectors $M_{\mathbb{Q}} \tilde{Q}_i$ and $M_{\mathbb{Q}}(\tilde{Q}_i - \tilde{Q}_j)$ in condition $(B_{\varepsilon}^{\mathbb{Q}})$. It uses $71/50$ and $56/25$ in place of $\sqrt{2}$ and $\sqrt{5}$. Under these changes both theorems remain valid.*

Sketch of proof. By Lemma 6.1, s and c satisfy the bound $\kappa/7$ of [SY25, Lemma 38]. That bound is the only property of $\sin_{\mathbb{Q}}$ and $\cos_{\mathbb{Q}}$ used in [SY25, Lemma 40 and Corollary 41], so their matrix bounds hold for the matrices built from s and c . In [SY25, Lemmas 44, 46 and 49 and Corollaries 50 and 51] each stated bound exceeds the bound derived in its proof by more than 0.9κ . For example, $\|MP - M_{\mathbb{Q}}\tilde{P}\| \leq 2\kappa(1 + \kappa)^2$ is stated as 3κ , and $3\kappa(1 + \kappa)^3$ as 4κ . Rounding a vector entry to 10^{-13} changes an inner product with a vector of norm at most $2(1 + \kappa)$ by less than 4×10^{-13} , and that margin absorbs it. We expect the second-derivative terms of [RR] to follow the same pattern, but we have not checked this in the Lean source. Finally, $71/50 > \sqrt{2}$ and $56/25 > \sqrt{5}$, and each occurrence lies where a larger value makes the hypothesis harder to satisfy. In $(B_{\varepsilon}^{\mathbb{Q}})$ a larger denominator does so because the condition forces the numerator to be positive. $\square$

This sketch is one of the points that an independent review must confirm (Section 7.5).

7. THE CERTIFICATE

7.1. The cover. The certificate covers the root box B_0 of Corollary 2.4. A fixed grid of $4 \times 30 \times 4 \times 15 \times 30 = 216,000$ cells divides the axes $\theta_1, \varphi_1, \theta_2, \varphi_2, \alpha$. All box endpoints are integers over $N = 15,360,000 \cdot 2^{12} = 62,914,560,000$, which is 2^{12} times the denominator used for the Noperthedron [SY25, Section 7.2]. Each cell holds a binary tree. A node either splits its box along one axis into two exact halves, or it is a leaf of one of four types:

- (G) the rational global theorem of [RR];
- (L) the rational local theorem [SY25, Theorem 48];
- (F) the rational gap form, Theorem 6.5;

(P) the rational patch form, Theorem 6.6.

A floating-point generator chooses the splits and the witnesses. Near the diagonal it splits the outer view first, until the outer view is 32 times narrower than the inner view. The local theorems tolerate a wide inner view, because their drift depends only on the outer view.

7.2. The replay. An exact checker, written in Python with integer and rational arithmetic, replays every cell. It confirms that the root of each tree is its grid cell, that each split yields the two exact halves, that every node is reached exactly once, and that every leaf passes its check. The generator and its database are not trusted. Table 2 gives the result.

Node type	Count
Split	1,593,149
Global leaf (G)	1,528,751
Published local leaf (L)	243,928
Gap-form leaf (F)	29,210
Patch-form leaf (P)	7,260
Total	3,402,298

TABLE 2. The nodes of the certificate. All 216,000 cells pass the exact replay. For comparison, the Noperthedron’s tree has 585,200 interior nodes and 18,115,245 leaves [SY25, Section 7].

The generator took 2.5 processor-hours and the replay 17.7, both with eight workers on an Apple M1 Max laptop. The first replay rejected seven cells. In each, the rational level τ of some gap-form leaf sat between 1.3×10^{-15} and 2.7×10^{-14} below an exact gap G_{kn} . The floating-point generator had padded τ by about 10^{-15} , while the rounded trigonometry moves the gaps by up to 3×10^{-14} . The checker rejected these leaves, as it should. The generator now checks every new local witness exactly before it accepts one. Five of the cells were rebuilt, and with the patch form available their trees shrank from 11,051 nodes in total to 1,183. In the other two, cells 119894 and 119895, the one failing leaf was regrown as a patch leaf. These two cells contain the hardest point of the band described below. The second replay accepted all 216,000 cells.

7.3. Where the new theorems are needed. Table 3 compares the largest diagonal box that each local theorem gives at sampled views. Most of the Undecanope’s diagonal is as easy as the Noperthedron’s. The hard views form one band near $\varphi \approx 0.725$ with $\theta \equiv 0.306$ modulo $2\pi/11$, together with its mirror image (Figure 2). There an antipodal pair of vertices crosses a silhouette edge. Without that pair the first-order depth falls to about 5×10^{-4} . The two hardest cells, 119894 and 119895, cover $\theta_1, \theta_2 \in [0.286, 0.429]$ and $\varphi_1, \varphi_2 \in [0.63, 0.74]$ near $\alpha = 0$, and they need 10,863 and 18,441 nodes.

7.4. Tests. The following tests are evidence, and they are not part of the proof.

Gap form. A test checked every inequality in the proof of Theorem 4.2 against the true shadows at 200 random poses in each of 406 diagonal boxes, half of them along the weakest first-order direction. No inequality failed, and the smallest slack was $+1.0 \times 10^{-9}$. In 370 off-centre boxes with unequal radii, 22,200 poses gave no failure, with smallest slack $+1.0 \times 10^{-7}$.

Patch form. A test checked every step of the proof of Theorem 6.6 at 631,544 poses inside the certificate’s own 7,260 patch leaves. It used 348,480 uniform poses, 232,320 box corners, 7,184 poses in segments with a positive gap, and 43,560 poses from bounded searches that

Largest diagonal box half-width	Minimum	0.1%	1%	Median
Undecanope, new theorem	3.5×10^{-4}	2.1×10^{-3}	4.2×10^{-3}	0.017
Undecanope, published theorem	fails	fails	2.9×10^{-3}	0.042
Undecanope, better of the two	3.5×10^{-4}	4.2×10^{-3}	8.3×10^{-3}	0.042
Noperthedron, better of the two	4.2×10^{-3}	7.7×10^{-3}	0.012	0.040

TABLE 3. Quantiles of the largest box about a diagonal pose that each local theorem closes, over 16,384 Sobol views (seed 20261103), in floating point. The published theorem fails at 41 of the Undecanope’s views. The new theorem gives a positive box at every view.

minimise each margin divided by ψ . No step failed. The smallest margins were $2.0 \times 10^{-4}\psi$ for the main inequality and $2.5 \times 10^{-4}\psi$ for the exclusion, and the largest containment scale was $1 - 1.4 \times 10^{-6}$.

Negative controls. Three deliberate errors, each on 40 leaves, show what the patch test can catch. A vertex index shifted by one broke six steps at almost every pose. A halved $\psi_{\max}$ broke Lemma 3.1(c). Dropping e from the loss broke the Lipschitz step at 2 of 3,320 poses, while the main inequality still held. So the test checks each step on its own.

Attacks. The search attacked the solid globally in each of its four training rounds, and an independent audit ran three more restarts. Every attack found only the diagonal, at scale one. Two targeted searches then attacked the hard band. One stepped along the weakest first-order direction and climbed in all five pose coordinates. The other started from 400 views around the band, at rotations from 10^{-4} to 10^{-2} . Every climb returned to the diagonal at scale one.

7.5. What the proof trusts. The proof of Theorem 1.1 rests on three things. The first is the mathematics: Sections 2 to 6, and the theorems of [SY25] and [RR]. The second is the exact checker, whose files the manifest lists with their SHA-256 digests (Appendix A). Two of those files also hold the floating-point witness search, which the checker does not use. The third is Python’s integer and rational arithmetic. The following steps remain before the result can be called verified.

- (1) An independent review of Sections 3 to 6, including Lemma 6.7.
- (2) A second implementation of the two new leaf checks, written from Theorems 6.5 and 6.6 alone, and a replay of their 36,470 leaves.
- (3) A formalisation of the new theorems and the finite certificate in Lean, building on [RR].

8. OPEN QUESTIONS

Question 8.1. The odd-cyclic family contains non-Rupert solids with 90 vertices (C_{15} , three orbits) and 88 vertices (C_{11} , four orbits). What is the least number of vertices of a non-Rupert solid in this family?

Question 8.2. Steininger and Yurkevich conjectured that the rhombicosidodecahedron is not Rupert [SY23, Conjecture 2], and their local theorem fails at some of its views [SY25, Section 9.1]. At its top view several vertices project to the same point, so the first shell of Corollary 4.4 must come from other vertices. Do the gap and patch forms close these views?

Question 8.3. The drift e of Lemma 4.3 uses one radius for both outer angles. A version with a separate radius for each angle, as [RR] has for the global theorem, would shrink the trees. How much?

Question 8.4. The first-order analysis uses point symmetry, which removes the translation. Can it be extended to solids without point symmetry, such as the stellated tetrahedra of [Zen26], where a translation enters the first-order model?

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AI models did most of the computation and mathematics in this paper, under the author's direction. The work started in OpenAI's Codex and finished in Anthropic's Claude Code. GPT-5.6 Sol, in Codex, wrote the search for odd-cyclic candidates and the shared exact engine for the published theorems. Its search produced the coordinates of the Undecanope. Claude Opus 5.5, in Claude Code, then wrote the first-order test. It found and proved the results of Sections 3 to 6, wrote their exact checks and tests, built and replayed the certificate, and drafted this paper. The author chose the problem, directed the work and is responsible for the claims made here. No one outside this project has yet reviewed the new theorems, and the author welcomes corrections.

DATA AVAILABILITY

The code, the frozen vertex tables, the certificate database, the replay results and the test records are deposited at Zenodo [Data]. The certificate database holds the 3,402,298 nodes in 108 MB compressed.

APPENDIX A. DATA FOR THE UNDECANOPE

Vertex $k + 11j + 44\ell$, for $0 \leq k \leq 10$, $0 \leq j \leq 3$ and $\ell \in \{0, 1\}$, is $(-1)^\ell R_z(2\pi k/11) g_{j+1}$. The vertex table holds $[10^{16}P]$ for each vertex P , and the SHA-256 digest of its canonical JSON form is

5e5412b6c88a231385e7957d6710f3a9a042d8ea3c952c711321c29753a847f9.

The largest squared approximation error is below $103/(5 \cdot 10^{33})$. The manifest records the SHA-256 digest of the checker files,

fd7b7a33db04c0c629577253d27278bdec14bb0621e7829e74ffe252995c7cc3,

and the digest of the logical tree,

80af76b03b27edec111ee3384e34b36db65e8d430bea8413498a298f8a9feb8f.

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